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UNSUPERVISED LEARNING

Task– Discover hidden patterns or structures in the data to potentially reveal patterns like diabetes risk levels or other health conditions.



Experience— Suppose we have a dataset of individuals with the following columns:

- ▶ x_1 : Fasting blood sugar level
- ▶ x_2 : Body Mass Index (*BMI*)
- ▶ Unlike supervised learning, unsupervised learning does not use labeled outcomes like diabetes development variable.

Formal Presentation– We represent our dataset as a collection of pairs: $\{x^{(1)}, x^{(2)}, \dots, x^{(n)}\}$, where:

► $x^{(i)} = [x_1^{(i)}, x_2^{(i)}]$: is a vector representing the health indicators of the i^{th} individual.

Performance— In unsupervised learning, measuring performance is often more challenging due to the absence of a ground truth for comparison. More on this in future lectures!



[90,22.5]



[180,28.7]



[130,25]



[145, 27.5]



Task Discover hidden patterns or structures in the data to potentially reveal

patients like diabetes risk levels or the health conditions.

like diabetes dependent variable.



Experience—Supervise what we have and as set of individuals with the following

:Fasting blood sugar level

► Unlike supervised learning, unsupervised learning does not use labeled outcomes

:BodyMassIndex(

x

1

x2

BNI

Performance-In supervised learning, measuring performance is often more challenging due to the absence

of a ground truth for comparison. More on this in future lectures!

clusters of individuals with

similarities





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3



[90,22.5]



[180,28.7]



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[145, 27.5]